

CWT-KWL series PT100 module manual



Model list

Model	Parameter
CWT-KWL-2PT	2 channels PT100, RS485 output
CWT-KWL-4PT	4 channels PT100, RS485 output
CWT-KWL-8PT	8 channels PT100, RS485 output
CWT-KWL-16PT	16 channels PT100, RS485 output

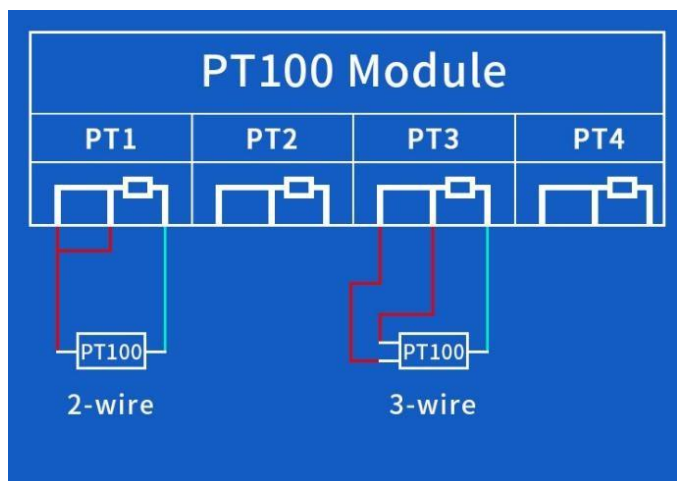
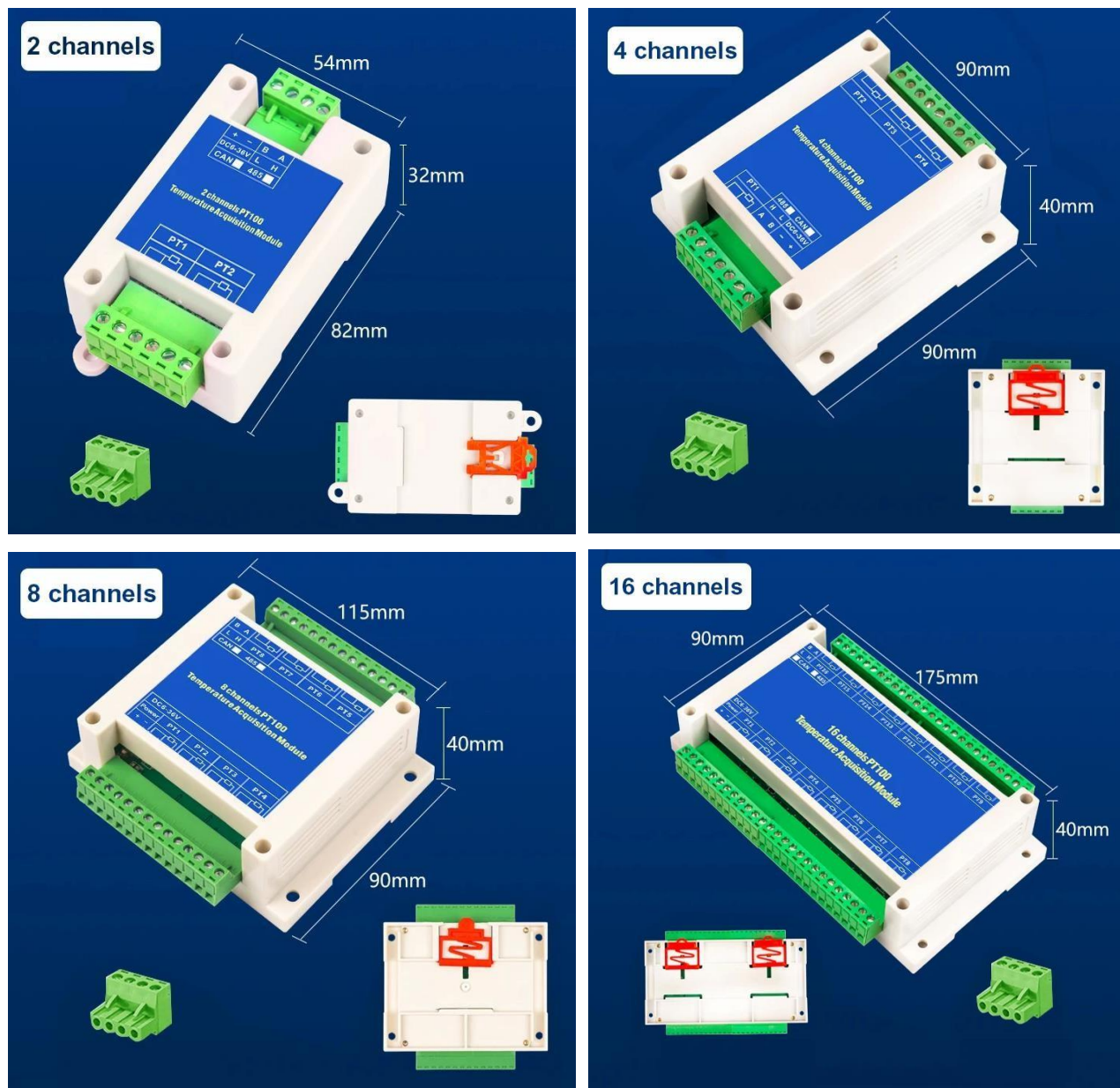
Specification

PT100 module specification:			
Supply power	6-36V DC	communication port	Isolated RS485
Number of channels	1-16 channel	Temperature range	-200~850℃
Power consumption	<200mW	sensor type	2 or 3 wires
Temperature accuracy	±0.3℃	Resolution	0.1℃
Acquisition rate	100KHZ/channel	Data Update	10HZ/channel
Error alarm	Sensor disconnects detection		
baud rate	1200~115200 (default 9600)		

Characteristic

- Double watchdog of software and hardware, never down
- Read and use, no calculations required
- Sensor disconnects detection
- Overvoltage protection, overcurrent protection
- Simple connection, plug and play
- All channels integrated into one output
- Isolated communication, strong anti-interference
- Industrial grade quality, reverse polarity surge protection
- Digital signal output to prevent acquisition errors
- Wide measuring range, high measuring accuracy

Size



RS485 communication (Modbus RTU protocol)

Default parameters: 9600,n,8,1

Modbus register map

Address (hex)	PLC address	Description	Fcode	Scale	data type	Property
00	40001	Channel 1 temperature	03	0.1	Int16	R
01	40002	Channel 2 temperature	03	0.1	Int16	R
02	40003	Channel 3 temperature	03	0.1	Int16	R
03	40004	Channel 4 temperature	03	0.1	Int16	R
04	40005	Channel 5 temperature	03	0.1	Int16	R
05	40006	Channel 6 temperature	03	0.1	Int16	R
06	40007	Channel 7 temperature	03	0.1	Int16	R
07	40008	Channel 8 temperature	03	0.1	Int16	R
08	40009	Channel 9 temperature	03	0.1	Int16	R
09	40010	Channel 10 temperature	03	0.1	Int16	R
0A	40011	Channel 11 temperature	03	0.1	Int16	R
0B	40012	Channel 12 temperature	03	0.1	Int16	R
0C	40013	Channel 13 temperature	03	0.1	Int16	R
0D	40014	Channel 14 temperature	03	0.1	Int16	R
0E	40015	Channel 15 temperature	03	0.1	Int16	R
0F	40016	Channel 16 temperature	03	0.1	Int16	R
14	40021	Device ID (1-247, default is 1)	03/06	1	uint16	R/W
15	40022	Baud rate: 0: 1200 1: 2400 2: 4800 3: 9600 (default) 4: 19200 5: 38400 6: 57600 7: 115200	03/06	1	uint16	R/W
16	40023	Channel 1 temperature offset	03/06	0.1	Int16	R/W
17	40024	Channel 2 temperature offset	03/06	0.1	Int16	R/W
18	40025	Channel 3 temperature offset	03/06	0.1	Int16	R/W
19	40026	Channel 4 temperature offset	03/06	0.1	Int16	R/W
1A	40027	Channel 5 temperature offset	03/06	0.1	Int16	R/W
1B	40028	Channel 6 temperature offset	03/06	0.1	Int16	R/W
1C	40029	Channel 7 temperature offset	03/06	0.1	Int16	R/W
1D	40030	Channel 8 temperature offset	03/06	0.1	Int16	R/W
1E	40031	Channel 9 temperature offset	03/06	0.1	Int16	R/W
1F	40032	Channel 10 temperature offset	03/06	0.1	Int16	R/W
20	40033	Channel 11 temperature offset	03/06	0.1	Int16	R/W
21	40034	Channel 12 temperature offset	03/06	0.1	Int16	R/W
22	40035	Channel 13 temperature offset	03/06	0.1	Int16	R/W
23	40036	Channel 14 temperature offset	03/06	0.1	Int16	R/W
24	40037	Channel 15 temperature offset	03/06	0.1	Int16	R/W
25	40038	Channel 16 temperature offset	03/06	0.1	Int16	R/W

Note: temperature register value= -5000(DEC) mean PT100 is not connected or faulty
When temperature less than 0, value will be responded in complement.

Read temperature

For example, read channel 1-3 temperature:

Master sends

Address	Function Code	Start Address (Hi)	Start Address (Lo)	Number of Points (Hi)	Number of Points (Lo)	Error Check (Lo)	Error Check (Hi)
0x01	0x03	0x00	0x00	0x00	0x03	0x05	0xCB

Module responds:

Address	Function Code	Number of byte	Channel 1 temperature	Channel 2 temperature	Channel 3 temperature	Error Check (Lo)	Error Check (Hi)
0x01	0x03	0x06	0x00 0xB2	0x00 0xB5	0xEC 0x78	0x44	0x6B

Channel 1 temperature = 00B2 H = 178 => 17.8°C

Channel 2 temperature = 00B5 H = 181 => 18.1°C

Channel 3 temperature = EC78 H = -5000, PT100 is not connected or faulty

Set slave ID

E.g., set slave ID=2, Master sends

Address	Function Code	Start Address (Hi)	Start Address (Lo)	ID	Error Check (Lo)	Error Check (Hi)
0xFE	0x06	0x00	0x14	0x00 0x02	0x5C	0x00

Module responds:

Address	Function Code	Start Address (Hi)	Start Address (Lo)	ID	Error Check (Lo)	Error Check (Hi)
0xFE	0x06	0x00	0x14	0x00 0x02	0x5C	0x00

Note: FE is broadcast address

Enquiry slave ID

Master sends

Address	Function Code	Start Address (Hi)	Start Address (Lo)	Number of Points (Hi)	Number of Points (Lo)	Error Check (Lo)	Error Check (Hi)
0xFE	0x03	0x00	0x14	0x00	0x01	0xD0	0x01

Module responds:

Address	Function Code	Number of Points	address	Error Check (Lo)	Error Check (Hi)
0xFE	0x03	0x02	0x00 0x01	0x6D	0x90

Set baud rate

E.g., set baud rate to 4800, Master sends

Address	Function Code	Start Address (Hi)	Start Address (Lo)	command	Error Check (Lo)	Error Check (Hi)
0xFE	0x06	0x00	0x15	0x00 0x02	0x0D	0xC0

Module responds:

Address	Function Code	Start Address (Hi)	Start Address (Lo)	command	Error Check (Lo)	Error Check (Hi)
0xFE	0x06	0x00	0x15	0x00 0x02	0x0D	0xC0

set baud rate to 9600, Master sends

Address	Function Code	Start Address (Hi)	Start Address (Lo)	command	Error Check (Lo)	Error Check (Hi)
0xFE	0x06	0x00	0x15	0x00 0x03	0xCC	0x00

Module responds:

Address	Function Code	Start Address (Hi)	Start Address (Lo)	command	Error Check (Lo)	Error Check (Hi)
0xFE	0x06	0x00	0x15	0x00 0x03	0xCC	0x00